

Fortgeschrittenenpraktikum

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HS1 Radio Astronomy

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Abstract

As part of “Fortgeschrittenenpraktikum” of the University of Hamburg this experiment’s procedure, in the period March 4 - 8, 2024, includes measuring radio signals of the sun with the aim of characterizing a radio telescope and determining the rotational velocity and the mass of the Milky Way with help of the spectra of the 21 cm line.

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1 Introduction

The experiment “radio astronomy” aims to familiarize the participants with basic radio astromonic methods of observation, especially those using individual antennas. For this experiment in particular, the radio telescope 300 A is used, which is located on top of the main building of the “Hamburger Sternwarte” in Bergedorf. Initially, the telescope must be calibrated and characterized in preparation for the measurement of radio signals from the sun, with the aim of measuring the spectra of the 21 cm hydrogen line of the Milky Way, which forms the main part of the experiment.

The protocol can therefore be roughly divided into 5 parts: (1) Introduction with theoretical background, (2) Observation and evaluation, (3) Results, (4) Discussion and (5) Appendix with references and list of figures. The second part can be further subdivided into (2.1) Calibration, (2.2) Determination of the angular resolution, (2.3) Determination of the sensitivity and (2.4) Observation of the 21 cm hydrogen line. In addition, at the beginning of part (2) a brief overview of weather data, experiment setup and program settings is given.

1.1 The 21 cm radiation of neutral hydrogen atom

In the Milky Way there are so-called HI regions in the interstellar space. These are regions that are rich in neutral hydrogen. The 21 cm radiation can emit from these regions.

The 21 cm radiation is due to the hyperfine structure of neutral hydrogen in the ground state. When the electron and proton spins align parallel, the hydrogen atom occupies a higher energy state, compared to the anti parallel alignment, that corresponds to a lower energy state. The transition of these states emits a photon with a wavelength of 21 cm (frequency of 1420 MHz). This transition has a transition rate in the range of 10^{-15} s^{-1} and a very long lifetime of about 11 Myr. Due to the low density of the HI regions, there are few collisions between the atoms, so that the excited state is occupied. In addition, this process occurs a lot in the universe, and therefore also in the Milky Way, which makes the occurrence of such radiation measurable.

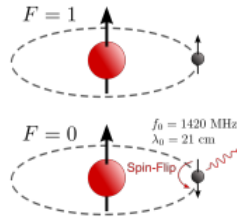


Figure 1: The flip of the spin results in the 21 cm radiation. Image taken from [7].

1.2 Relativistic Doppler Effect

If you observe the hydrogen distribution in the Milky Way, you will notice that the 21 cm line is not measured exactly at 21 cm, but as a distribution. The 21 cm radiation undergoes a frequency shift due to the Doppler effect. When hydrogen moves away from an observer, the received light shifts towards the higher wavelengths. Conversely, when hydrogen approaches the observer, the shift is towards lower wavelengths. This phenomenon arises because light experiences a change in energy when emitted by a moving object. As a photon travels in the same direction as the emitting object, it gains energy, leading to an increase in its total energy and consequently, its frequency. However, since the speed of light remains constant across all frames of reference, this energy increment does

not translate into additional speed. Conversely, when a photon moves in the opposite direction to the emitting object, it loses energy and consequently, its frequency decreases. This phenomenon is called the relativistic Doppler effect and is described by the following formula:

$$f_B = f_s \cdot \sqrt{\frac{1 + \beta}{1 - \beta}}, \quad (1)$$

where f_B is the frequency observed, f_s the frequency which is being sent out by the moving object and $\beta = v/c$ with v being the velocity of the object relative to the observer.

1.3 Differential rotation in the Milky Way

In order to make measurements of the movements in the Milky Way, one must take into account that these observations take place in the plane of the Milky Way itself. When observing objects in the galaxy within the galaxy itself, you only see the line of sight projection of the velocity vector from the object. Figure 2 shows a schematic representation of the circumstances. The actual rotational velocity can be derived from this in the following.

In general, the maximum speed of an object can be observed when the line of sight to the object is tangential to its trajectory:

$$v_{\max} = v_{\text{obs}} - v_{\text{LSR}},$$

where v_{LSR} is the local standard of rest velocity, i.e., a correction due to the movement of the measuring point. In 2.4 the maximum of v_{LSR} is measured, and the orbital velocity v_{rad} is calculated with

$$v_{\text{rad}} = \omega R - \omega_0 R_0 \sin l = v - v_\odot \sin l \quad \text{with} \quad R = R_\odot \sin l$$

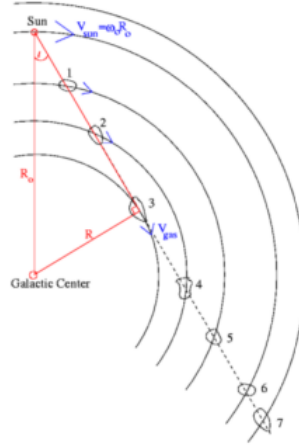


Figure 2: Schematic representation of the movement of mass in the Milky Way from [4].

1.4 Spatial resolution of single-dish telescopes and radio interferometers

Single-dish antennas offer limited spatial resolution due to the diffraction limit, which is determined by the telescope's wavelength and aperture size. The diffraction pattern from a circular aperture produces a central bright disc and fainter outer rings, with the angular diameter of the central disc defined by the equation $q = 2.44 \lambda/D$, where λ is the wavelength, D is the aperture diameter, and q is the angular diameter in radians [4]. In radio astronomy, resolution often refers to the half-width of the central disc (main lobe), representing the width at half of the intensity maximum (FWHM), according to the Rayleigh criterion for distinguishing separate objects. The Rayleigh

criterion says two light sources can just be distinguished when one's main bright spot aligns with the first minimum spot of the other's diffraction pattern. Therefore, the spatial resolution of a radio telescope is described as:

$$q = 1.02 \frac{\lambda}{D} \text{rad} = 58.4^\circ \frac{\lambda}{D}. \quad (2)$$

2 Observation and evaluation

For the sake of clarity, this section first gives an overview of the weather data, the setup itself, the computer program, the positioning accuracy, the pointing corrections and the BBC settings. Later in the subsections the respective experiments and observations are explained, which are subdivided into the parts 2.1 Calibration, 2.2 Determination of angular resolution, 2.3 Determination of sensitivity and 2.4 Observation of the 21 cm line.

The experiments in the following sections 2.1 to 2.3 were conducted on **March 4, 2024**. The weather was changeable that day, the temperature was between 2 and 14 degrees Celsius and the relative humidity was around 70 percent. There was no precipitation, but it was a little foggy in the morning and the sky was rather overcast during the day. The average wind speed was around 14 kmh^{-1} and the wind direction was north-east (see appendix figure 11).

The experiment in section 2.4 was started on **March 5, 2024**. The weather on this day was also changeable, the temperature was between 1 and 9 degrees Celsius, the relative humidity fluctuated between 50 and 90 percent. There was no precipitation and the sky was rather overcast until the afternoon hours. The average wind speed was around 18 kmh^{-1} and the wind direction was north-east (see appendix figure 11).

Section 2.4 was completed on **March 6, 2024**. The weather on this day was sunny, the temperature was between -1 and 9 degrees Celsius, the relative humidity fluctuated between 55 and 95 percent. There was no precipitation and the sky was clear until well into the afternoon. The wind speed averaged around 12 kmh^{-1} and the wind direction was north-east (see appendix figure 11).

For the following measurements, the SPIDER 300A radio telescope was used, which is located on the roof of the observatory in Bergedorf. It is a parabolic antenna, has a diameter of 3 meters and operates in a frequency range from 1398 MHz to 1462 MHz (see appendix figure 12). The telescope is operated using the RadioUniversePRO software, which can move the telescope horizontally and vertically and also offers various other measurement methods. The user interface is shown in figure 13. In this case, the "BBC Tools" tab is open, but ten other tabs can be opened as well: "IF Monitor", "Offset Alignments", "Source Visibilities", "Gain Calibration", "Obs Data", "OnOff", "Spectrometer", "Total Power Plots", "Mapping" and "Environmental Data". The associated measurement methods are explained in more detail at the respective points in the protocol. The receiver on the telescope, which consists of 16 Base Band Converters (BBC), i.e. 8 per polarization, measures left and right circularly polarized radio waves, which are forwarded to the software via a cable. As each band has a bandwidth of 8 MHz, a total of 64 MHz can be recorded per polarization. The bands can be switched on and off manually via the software so that, for example, only the two bands with a frequency of 1420 MHz can be set for the measurement of the 21 cm line in the main part of the measurement in section 2.4. The measurement data is given in counts, which is a non-calibrated dimensionless measurement of the total power (TP). The total power measured at the receiver can be increased using the gain setting so that a better signal can be recorded, whereby the TP counts should be around 20000 counts to avoid oversaturating the filter.

During the three measurement days, the receiver settings of the telescope, in particular the gain, and the positioning accuracy had to be readjusted every day. For the sake of clarity, these settings are listed together here.

Due to noise, only the upper side bands BBC 01 to 05 for the left polarization and BBC 09 to 13 for the right polarization could be selected during the entire measurement period. In order to achieve a maximum power (TP) of around 20000, the gain had to be adjusted. The constant setting of

+18 for the right and +18 for the left could be used for this over the three days. Furthermore, the exact position of the sun had to be determined each day as preparation before the measurements. This is due to the fact that although the program has a pre-calculated position of the sun, small deviations can sometimes occur. For such a measurement, a procedure can be carried out using the Offset Alignment tab (see appendix figure 18(a) for the settings) in which the telescope scans the sky over an offset position in the form of a cross and records the measured power at the receiver. The point of highest power which is determined with a Gauss fit is determined as the point of the sun. If the positioning is accurate, a well-defined maximum should be visible at an offset of 0 degrees longitude and latitude. This was approximately the case on all three days, so that no further pointing correction had to be made over the experiment period. An example of the power as a function of the longitude and latitude can be found in the appendix figure 14.

2.1 Calibration (04.03.2024)

Before the telescope is ready to take measurements and record data, it must first be calibrated. For this purpose, the so-called "On-Off measurement" and "Hot-Cold measurement" are carried out later. This is based on the theory described in the following.

For each measurement, the receiver detects radiation from the source T_s , radiation from the environment T_h and radiation from the receiver and amplifier T_{sys} which represents the pure system temperature. The measured power of the receiver can therefore be described by the antenna temperature T_A [1]:

$$T_A = T_s + T_h + T_{sys} \quad (3)$$

By assuming that the pure system temperature and the background radiation are difficult to differentiate and that the source does not reduce the background radiation because it only covers a small part of the sky, an effective Temperature T_{sys} can be defined [1]:

$$T'_{sys} = T_{sys} + T_h \quad (4)$$

To determine the contribution of a source T_s at a measured antenna temperature T_A , the effective temperature T_{sys} must be determined, which is the aim of the calibration and can be carried out by the "On-Off measurement". For this purpose, the antenna temperature T_A is measured once with a source of known flux density and once at an empty spot in the sky. Since it can be assumed that the radiation from the surroundings T_h is constant, the system temperature T_{sys} is then also known for measurements at other sky positions. In order to achieve greater accuracy, it must be taken into account that the background temperature T_h depends on the altitude. Since the KRT 3 can only observe the sun with a sufficiently high flux density for calibration and the sun is a variable source that is not well suited for calibration, a noise generator is used instead and a "Hot-Cold measurement" is carried out.

2.1.1 Determination of the system temperature

The KRT 3 is equipped with a noise generator that feeds a signal directly into the antenna horn. The calibration temperature $T_{cal} = 170K$ is the expected antenna temperature T_A , which is also observed when the noise generator is switched on. Thus, T_{sys} can be estimated from such a "Hot-Cold measurement". During the experiment, the flux density of the sun is measured and the calibration is checked by comparing the measured value with the solar flux published daily online. Assuming that the sun is a radio source, the following equations apply [1]:

$$T_b \Omega_s = T_A \Omega_A \quad (5)$$

$$A_e \Omega_A = \lambda^2 \quad (6)$$

$$B_\nu = \frac{2k_B T_b}{\lambda^2} \quad (7)$$

$$\eta = \frac{A_e}{A_g} = 0.5 \quad (8)$$

$$S_\nu = B_\nu \Omega_s \quad (9)$$

Here, T_b is the brightness temperature, Ω_s the solid angle of the sun, T_A the antenna temperature, Ω_A the solid angle of the antenna main lobe, A_e the effective antenna area, λ the wavelength, B_ν the spectral radiation, k_B the Boltzmann constant, η the aperture efficiency, A_g the geometric antenna area and S_ν the flux density. To convert the measured antenna temperature T_A into a flux density S_ν , the following calculation must be carried out first:

$$T_b \Omega_s = T_A \Omega_A = T_A \frac{\lambda^2}{A_e} \quad (10)$$

$$T_b \Omega_s = \frac{B_\nu \lambda^2}{2k_B} \cdot \Omega_s = S_\nu \frac{\lambda^2}{2k_B} \quad (11)$$

The two results for $T_b \Omega_s$ are now equated and transposed to T_A :

$$T_A \frac{\lambda^2}{A_e} = S_\nu \frac{\lambda^2}{2k_B} \quad (12)$$

$$T_A = \frac{S_\nu A_e}{2k_B} \quad (13)$$

In the following, an On-Off measurement with the sun as the source and a Hot-Cold measurement with a noise generator are carried out. Therefore, the telescope is pointed at the sun via choosing it in the "Source Visibility" tab. In the "BBC Tools" tab band 01 to 05 and 09 to 13 are selected due to strong interference for band 06 to 08 and 14 to 16. The first eight bands consider the left circular polarization and the last eight bands the right circular polarization. In the "On-Off" tab some setting are made before the On-Off measurement is started, which can be found in the appendix figure 18(b).

For the evaluation, the system temperatures $T'_{sys, left}$ and $T'_{sys, right}$ and the antenna temperature of the sun T_{sun} are calculated with the following equations (14), (15) [1] their averages and standard deviations (see appendix 15, 16, 17 for all calculated temperatures). The results are listed below:

$$T'_{sys} = T_{cal} \frac{P_{off}}{P_{cal} - P_{off}} \quad (14)$$

$$T_{sun} = T'_{sys} \frac{P_{on} - P_{off}}{P_{off}} \quad (15)$$

$$T'_{sys,left} = -1160.45 \text{ K} \pm 378.24 \text{ K}$$

$$T'_{sys,right} = 296.63 \text{ K} \pm 226.09 \text{ K}$$

$$T_{sun,left} = -2011.61 \text{ K} \pm 621.57 \text{ K}$$

$$T_{sun,right} = 2511.95 \text{ K} \pm 1141.88 \text{ K}$$

Since major measurement problems have occurred due to interference from the telescope receiver and the values for the left polarization are obviously inaccurate, only the values for the right polarization are used in the following for further calculations. Further sources of error are discussed at the end of the protocol in section 3.2.

Here P_{off} is the measured power in the OFF position, P_{cal} the measured power in the OFF position with the noise generator switched on and P_{on} the measured power in the ON position. The units of the measured powers are arbitrary.

Next, the theoretically expected antenna temperature of the sun T_A can be calculated with help of the previous equation (13) and the current flux density of the sun $S_\nu = 70 \text{ sfu} = 7 \cdot 10^{-22} \text{ W m}^{-2} \text{ Hz}^{-1} = 70000 \text{ Jy}$ [2]:

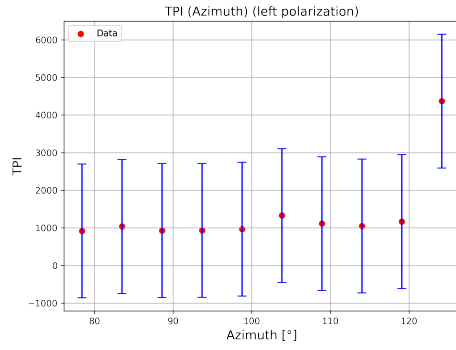
$$T_A = \frac{S_\nu A_e}{2k_B} = \frac{0.5\pi S_\nu (1.5)^2}{2k_B} = 896.38 \text{ K} \quad (16)$$

Compared to the observed values $T_{A,left} = -2011.61 \text{ K} \pm 621.57 \text{ K}$ and $T_{A,right} = 2511.95 \text{ K} \pm 1141.88 \text{ K}$, the theoretically calculated value is significantly smaller, which could be a result of inaccuracies of the measured data that is discussed later in section 3.2.

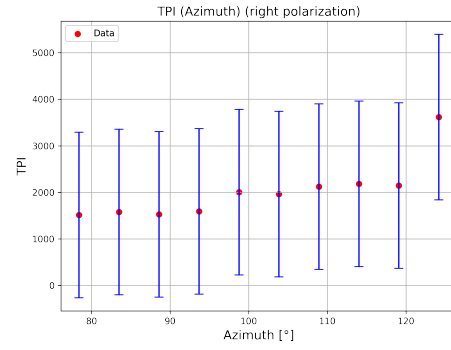
2.1.2 Background radiation

Due to the fact that there is no radio source in the antenna's visibility field at the OFF position, it can be approximated that $T_A \approx T'_{sys}$. For the examination of the background radiation the antenna temperatures T_A are measured at different azimuths which are 5 degrees apart and at different elevations (ELV) with a fixed azimuth. This cross-scan is performed over the supernova remnant TAU A (ELV = 44.24°), which is considered as a weak source and thus hardly stands out from the background. Therefore the telescope is pointed at TAU A by choosing it in the tab "Source Visibilities". The settings in the "BBC Tools" are the ones described at the beginning of section 2. For the cross-scan the span has to be calculated due to the fact that half a span is observed above and below the source and the minimum elevation has to be slightly above 10 degrees. Therefore, the span is $2(\text{ELV} - 10^\circ) = 68.48^\circ$. The settings for the tab "Total Power Plots" can be found in the appendix figure 18(c).

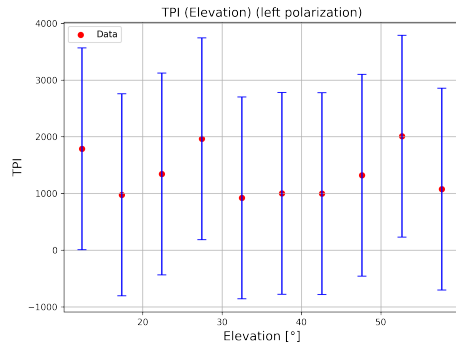
For the evaluation, the Total Power Intensity (TPI) is plotted as a function of offset in degrees in figure 3.



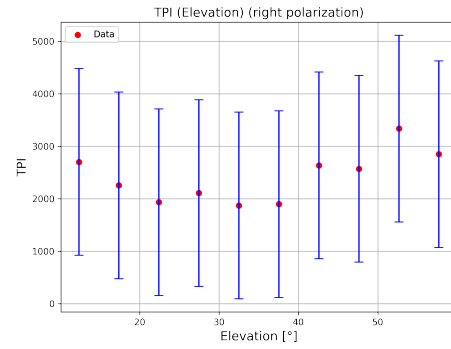
(a)



(b)



(c)



(d)

Figure 3: TPI as a function of azimuth [°] (pictures a and b) and elevation [°] (pictures c and d) for left and right polarization.

Moreover, the TPI is normalised at the horizontal scan to the calculated system temperature T'_{sys} so that the TPI can be normalised at the vertical scan as well. With these results T'_{sys} is plotted as a function of azimuth and elevation in figure 4).

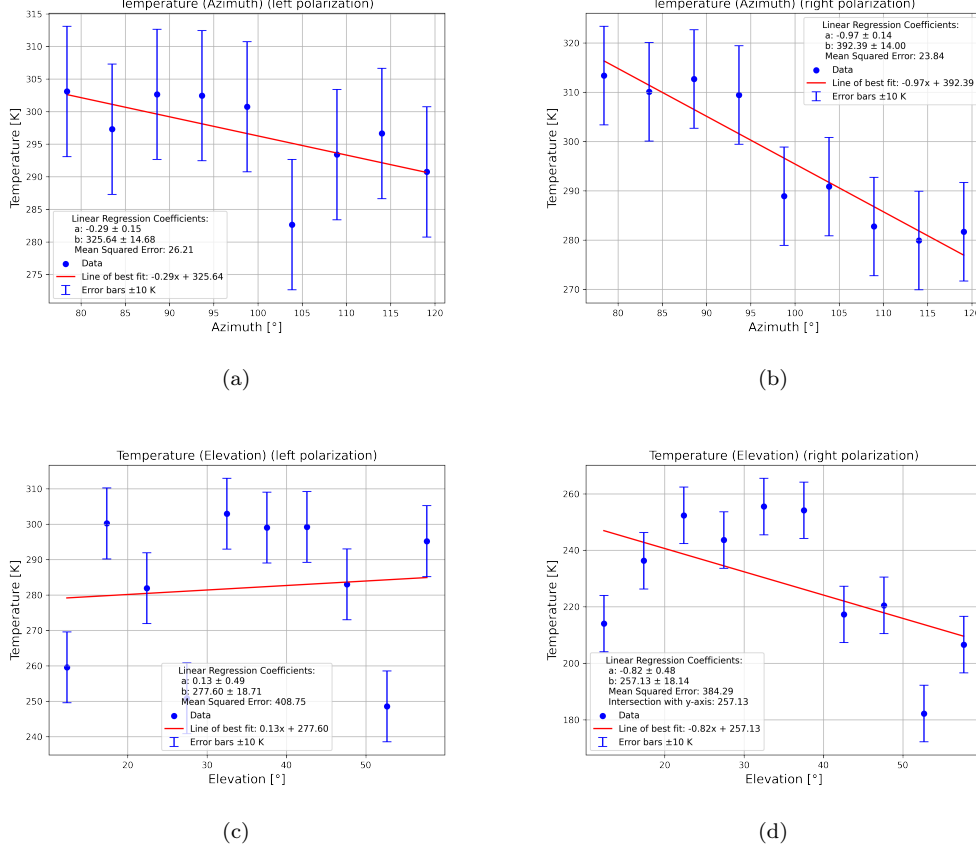


Figure 4: Cross scan over TAU A, showing the temperature [K] as a function of azimuth [°] (pictures a and b) and elevation [°] (pictures c and d) for left and right polarization. From that, the temperature at 0° can be determined.

Since the last data point of the measurement of the TPI as a function of azimuth appears to be an outlier, it is not considered further for normalization and extrapolation in order to keep the already poor data situation at least partially meaningful. Also due to the very poor data situation, an extrapolation up to an altitude of 0° can only be carried out with the right polarization data. Extrapolation is necessary here because the telescope cannot be set to an altitude of less than 10° . As can be seen from the plots, the intersection point for 0° is $257 \text{ K} \pm 18 \text{ K}$, which actually is the earth radiation temperature $T_{Earth} = 252 \text{ K}$. With regard to the temperature values in relation to the geographical latitude, it can be said that the closer the telescope points towards the Earth, the more it is influenced by its temperature, more precisely the blackbody radiation temperature of the Earth $T_{Earth} = 252 \text{ K}$ [4]. However, if the telescope is pointed towards the sky, the measured temperature consists of the background temperature T_h , which in this case represents the noise.

2.2 Determination of the angular resolution of the KRT 3 (04.03.2024)

To find the angular resolution, the telescope is pointed at the sun and the antenna temperature T_A is measured at various azimuth angles (in one-degree increments) and different elevations for both right and left circular polarization. In figure 5 the data is normalized to system temperature and four graphs are generated. The measurement data from the latitudinal right-polarized is unfortunately not usable for the evaluation for technical reasons.

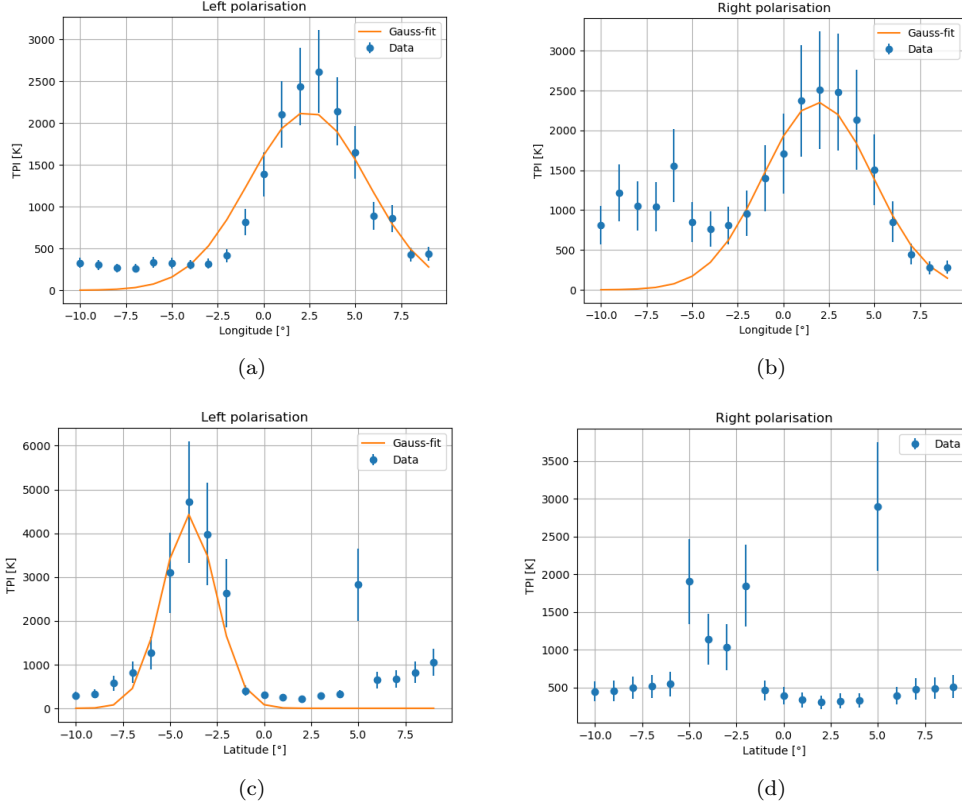


Figure 5: Cross scan of the sun with the received TPI in Kelvin on the y-axis and the offset from the sun in degrees on the x-axis. It is differentiated between left and right polarization as well as longitude and latitude.

In the plot of the right-polarised radiation of the longitude, a high degree of inaccuracy can be noted in the values from -10° to -3° . This may be due to interference that disturbs the signal. Similarly, in the latitude measurements, a systematic error can be recognised at 5° , which shows an uncharacteristically high value.

The longitudinal data for FWHM must be corrected by the factor $\cos(52^\circ)$, as it is not measured vertically upwards but in an angle. The FWHM from the remaining data can now be determined via $\text{FWHM} = 2\sqrt{2\ln(2)}\sigma$ and compared with the theoretical value. The values and errors of the parameters of the gaussian fit have been created by a python macro.

The average of the FWHM can be calculated from Table 1 as $4.13^\circ \pm 0.43^\circ$. The theoretically expected value for a wavelength of 21 cm and a diameter of 3m can be calculated as $q = 1.02 \cdot \lambda/D = 0.0714 \text{ rad} = 4.09^\circ$. The measured value is therefore very close to the theoretical expectations.

	Long Left	Long Right	Lat Left	Lat Right
μ	2.42	1.90	-3.98	/
Error σ_μ	0.47	0.53	0.14	/
σ	3.25	3.01	1.41	/
Error σ_σ	0.37	0.32	0.12	/
χ^2	1.44	0.62	1.35	/
FWHM in $[\circ]$	7.66	7.08	3.32	/
Error σ_{FWHM}	0.87	0.75	0.29	/
FWHM _{cor} in $[\circ]$	4.71	4.35	3.32	/
Error $\sigma_{FWHM_{cor}}$	0.53	0.46	0.29	/

Table 1: Measurement overview of Figure 5.

2.3 Determination of the sensitivity of the instrument (04.03.2024)

To determine the sensitivity of the instrument, first the theoretical sensitivity is calculated and subsequently the uncertainty of the radiant flux measurements is determined.

2.3.1 Theoretical sensitivity

For a given system temperature T'_{sys} the weakest detectable flux density ΔS_ν can be calculated with help of following radiometer equation in which ΔS_ν is the spectral density flux in $\text{W m}^{-2} \text{Hz}^{-1}$, k_B the Boltzmann constant, A_e the effective antenna surface, $C_s \approx 2$ a dimensionless constant, $\Delta\nu$ the bandwidth, τ the integration time and T'_{sys} the system temperature [1].

$$\Delta S_\nu = \frac{2k_B}{A_e} \frac{C_s T'_{sys}}{\sqrt{\Delta\nu\tau}} \quad (17)$$

First, for the evaluation the limiting sensitivity ΔS_ν in a single channel ($\Delta\nu = 61 \text{ kHz}$) for a case where 1 second is integrated is calculated using (17):

$$\Delta S_\nu = \frac{2k_B}{A_e} \frac{C_s T'_{sys}}{\sqrt{\Delta\nu\tau}} = 0.60 \cdot 10^{-23} \text{ W m}^{-2} \text{Hz}^{-1} \quad (18)$$

With a propagation of error:

$$\Delta(\Delta S_\nu) = \sqrt{\left(\frac{\delta(\Delta S_\nu)}{\delta T'_{sys}} \cdot \Delta T'_{sys}\right)^2} = 0.14 \cdot 10^{-23} \text{ W m}^{-2} \text{Hz}^{-1} \quad (19)$$

The error value is later evaluated in section 3.2. Next, the signal-to-noise ratio S/N is also calculated:

$$\frac{S_\nu}{\Delta S_\nu} = \frac{(940 \pm 50) \text{ Jy}}{(600 \pm 140) \text{ Jy}} = 1.57 \pm 0.48 \quad (20)$$

Then via proportion the integration time τ is calculated for Taurus A (ELV = 55.86°) for a signal-to-noise ratio = 10, $\nu = 1420.4 \text{ MHz}$ and a flux density $\Delta S_\nu = 940 \text{ Jy} \pm 70 \text{ Jy}$ [1]:

$$\tau = 2.46 \text{ s} \pm 0.23 \text{ s} \quad (21)$$

The error is also evaluated later in section 3.2. Since the signal-to-noise ratio is really poor [4] possible sources of errors are later described in section 3.2.

2.3.2 Determination of the uncertainty of radiant flux measurements

Measurements of the flux density are carried out relatively, so a measurement with the source (ON) is compared with a measurement without the source (OFF). Moreover, observation conditions can be considered constant, as the ON-OFF change occurs at very short intervals. To differentiate the background from the antenna temperature, a "beamswitch" measurement is taken, alternating between the position of the source (ON position) and a reference position next to the source (OFF position). The antenna temperature T_s can thus be defined:

$$T_s = (T_s + T'_{sys}) - T'_{sys} = T_A(ON) - T_A(OFF) \quad (22)$$

First, the flux density of a cell phone transmitting isotropically at a power of 2 W at 900 MHz on the earth and located on the moon is calculated. Therefore a normal distribution around 900 MHz with FWHM of 80 kHz is assumed ([1]). Using (17), $P = k_B T \cdot \Delta\nu$ ([4]) and $FWHM = 2\sigma\sqrt{2\ln 2}$ [5] following equations can be calculated:

$$\sigma = \frac{FWHM}{2\sqrt{2\ln 2}} = 33.93 \text{ kHz} \quad (23)$$

$$S_\nu = \frac{P}{4\pi d^2 \cdot \sqrt{2\pi}\sigma} = \frac{2 \text{ W}}{4\pi \cdot (384.8 \cdot 10^6 \text{ m})^2 \cdot \sqrt{2\pi} \cdot 33.93 \cdot 10^3 \text{ Hz}} = 1263.79 \text{ Jy} \quad (24)$$

Now, it is assumed that the spectral response is a simple power law of the form $S_\nu \propto \nu^\alpha$ with $\alpha = -0.7$. Knowing that the flux density of Cas A at 1420.4 MHz is $1420 \text{ Jy} \pm 70 \text{ Jy}$ ([1]) the flux density of Cas A at 900 MHz can be determined with help of a constant c to make the calculation more comprehensible:

$$\frac{(1420.4)^{-0.7} \text{ MHz}}{(1420 \pm 70) \text{ Jy}} = 4.38 \cdot 10^{-6} \pm 2.2 \cdot 10^{-7} \frac{\text{MHz}}{\text{Jy}} = c \frac{\text{MHz}}{\text{Jy}} \quad (25)$$

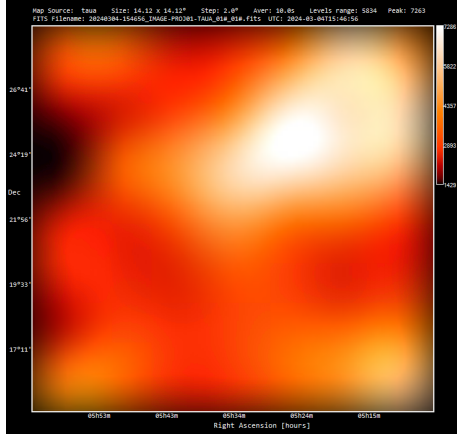
$$\frac{900^{-0.7}}{c} \text{ Jy} = \frac{900^{-0.7}}{4.38 \cdot 10^{-6} \pm 2.2 \cdot 10^{-7}} \text{ Jy} = (1954.37 \pm 98.31) \text{ Jy} \quad (26)$$

Therefore, for a frequency of 900 MHz the flux density of Cas A is $S_{\nu, CasA} = 1954.37 \text{ Jy} \pm 98.31 \text{ Jy}$ which is approximately 1.55 times the flux density of the mobile phone. This shows how strong the radio source of Cas A is, considering the enormous distance of 11000 ly or $1.04 \cdot 10^8 \text{ km}$ and that it is hardly comparable to the phone on the moon, which is only $3.84 \cdot 10^5 \text{ km}$ away.

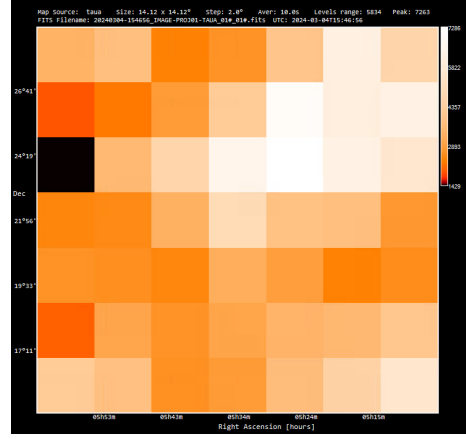
Now using the following equation ([4]), the antenna temperature T_A for the source Cas A can be calculated with the effective area of the antenna A_e , the flux density $\Delta S_\nu = (1420 \pm 70) \text{ Jy}$ ([1]), and the Boltzmann constant k_B :

$$T_A = \frac{\Delta S_\nu A_e}{2k_B} = 1.82 \text{ K} \pm 0.09 \text{ K} \quad (27)$$

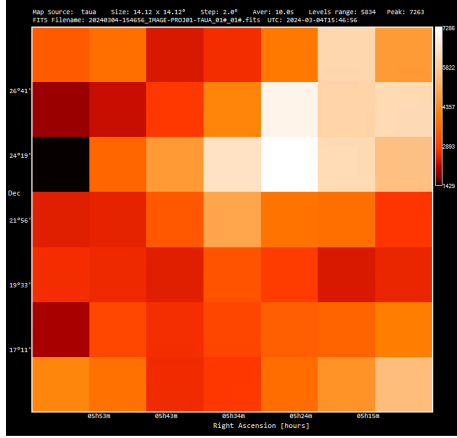
In following the antenna temperature of Tau A $T_{A, TauA}$ is determined by mapping it, which can be described as a pixel by pixel measurement of KRT 3. At the beginning of the observation Tau A has an elevation of 50.03° . The settings in the tab "Mapping" can be found in the appendix figure 18(d) and the mapping time is 32 min 01 s. Maps with different design settings are shown in figures 20(a), 20(b), 20(c) and 20(d).



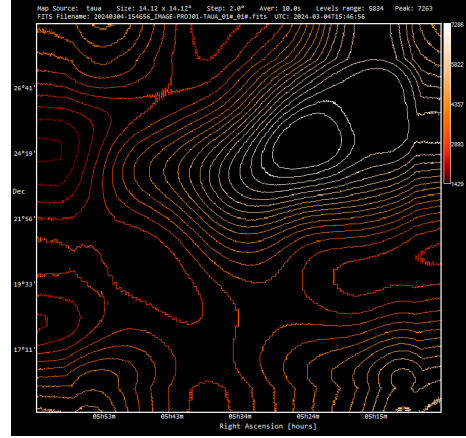
(a) Map with the settings "color = hot" and "square root".



(b) Map with the settings "pixelize", "color = hot" and "log".



(c) Map with the settings "pixelize", "color = hot" and "square root".



(d) Map with the settings "contour", "color = hot" and "square root".

Figure 6: Examples for maps with different settings.

The mean of the map is 158.55 and the standard deviation is ± 31.19 which is calculated by using the software ds9 for the measurement of the uncalibrated intensity values and a python macro to compute the mean value and standard deviation of the data. This allows the mean value to be normalized to the system temperature $T'_{sys} = 296.63 \text{ K} \pm 226.09 \text{ K}$ which also is $T_A(OFF)$. With this scaling done the map noise can be determined in units of Kelvin:

$$T_A(OFF) = T'_{sys} = 158.55 \pm 31.19 \triangleq 296.63 \text{ K} \pm 226.09 \text{ K}$$

Then, with the antenna temperature $T_A(ON)$ for Tau A can be determined with scaling which results in $T_{A,TauA}(ON) = 302.29 \text{ K} \pm 239.18 \text{ K}$. Moreover the antenna temperature of the source T_s can be calculated with the previous shown equation 22:

$$T_s = T_A(ON) - T_A(OFF) = (302.29 \pm 239.18)\text{K} - (296.63 \pm 226.09)\text{K} = (5.66 \pm 465.27)\text{K} \quad (28)$$

The background is not constant since there are effects of interference probably caused by the Milky Way. To compare this quantitatively the signal to noise ration S/N is calculated:

$$\frac{S}{N} = \frac{T_A(ON)}{T_A(OFF)} = \frac{(302.29 \pm 239.18)\text{K}}{(296.63 \pm 226.09)\text{K}} \approx 1.02 \pm 3.9 \quad (29)$$

This value is very similar to the theoretically calculated one for Cas A. Nevertheless, values smaller than 5 are considerate as not good (see reference [4]).

The actual measurement times can be calculated with the log. If 10 seconds are integrated per measurement point the percentage of measurement time, which is accounted for by the observation of the source, can be calculated:

$$\frac{(t_{tot} - t_{meas})}{t_{tot}} = \frac{32 \text{ min} \cdot 60 \text{ s} - 10 \text{ s} \cdot 49}{32 \text{ min} \cdot 60 \text{ s}} = 74.47\% \quad (30)$$

To detect the source with a signal-to-noise ratio $S/N = 10$, there are 32 ± 5 maps needed. Therefore the ideal case is assumed where $S/N \propto \sqrt{\tau}$ holds. In summary, the mapping is suitable for more imprecise observations, as the measurement can be considered good from a value upwards of $S/N = 5$ [4]. The quality of the measurement is most significantly distorted by external sources of interference and depends heavily on the integration time and the pixel resolution of the map.

2.4 Observation of neutral hydrogen in the Milky Way (05.03.2024 - 06.03.2024)

The aim of this experiment is to use the measurement of the frequency from the HI line to determine the maximum velocity of a HI cloud using the Doppler effect. From this the rotational velocity and finally the mass distribution of the Milky Way can be deduced. To do this, a few preliminary considerations must be made.

2.4.1 Determination of the frequency setting of the receiver

First, the velocity resolution δv from the frequency resolution $\delta \nu$ has to be derived. The following applies based on [1]:

$$\frac{\delta v}{c} = \frac{\delta \nu}{\nu_0} \quad \text{and} \quad \delta \nu = \frac{\Delta \nu}{1024}.$$

This determines the bandwidth $\Delta \nu = 1460.56 \text{ MHz} - 1398 \text{ MHz} = 62.56 \text{ MHz}$, since the preamplifier has a frequency range of $1398.00 \text{ MHz} - 1460.56 \text{ MHz}$, divided into 1024 channels. Thus, frequency resolution can be derived as $\delta \nu = 0.0611 \text{ MHz}$ and thus:

$$\delta v = c \cdot \frac{\delta \nu}{\nu_0} = 12.9 \text{ kms}^{-1}.$$

The HI line can be expected at a radial velocity of 0 km s^{-1} in the BBC channels 3 and 11. Assuming a maximum radial velocity of $\pm 200 \text{ km s}^{-1}$. This leads to a frequency shift of 0.94 MHz . A channel has a frequency range of 8 MHz , so BBC channels 3 and 11 are sufficient to measure the HI line.

2.4.2 Measurement of the 21 cm line along the Milky Way plane

In the following measurements, the spectrum of the plane of the galaxy is measured, changing the galactic longitude from 20° to 90° in steps of 10° . As shown in figure 7, the generated spectrum of the HI line can be recognised. Further plots can be found in the appendix 19, 20. Table 2 shows the relevant data as a table. The galactic longitude has an error of $\pm 1^\circ$.

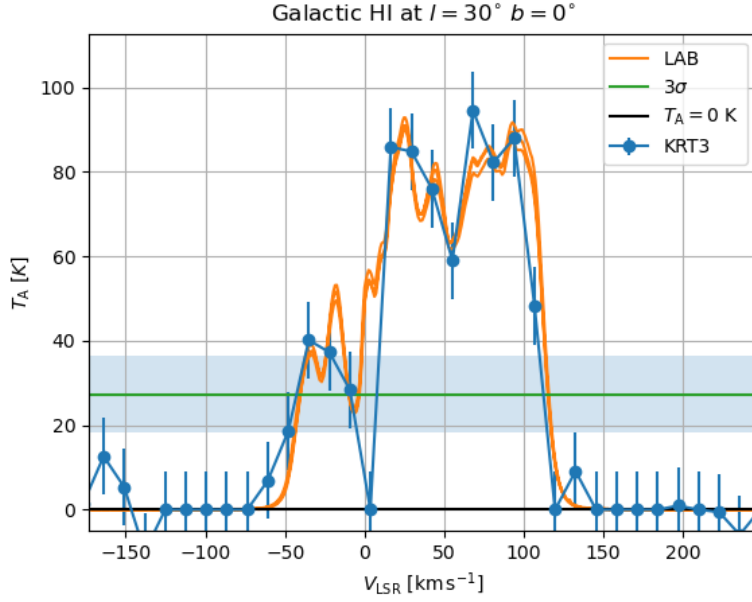


Figure 7: An example spectrum of the recorded 21 cm line, depicting the Radial velocity at a galactic longitude of 30° , is illustrated in blue. The comparison spectrum from the University Bonn is represented in orange [6].

Gal. long. [$^\circ$]	Radius [kpc]	Radius error [kpc]	v_{max} [km s^{-1}]	v_{max} error [km s^{-1}]	v_{rot} [km s^{-1}]	v_{rot} error [km s^{-1}]
20	2.9	0.14	110	18.43	185.244	22.13
30	4.25	0.13	112.1	9.2	222.09	12.5
40	5.46	0.11	85.3	7.9	226.711	10.8
50	6.51	0.10	72.0	5.6	240.5	8.1
60	7.36	0.08	35.75	23.4	204.27	25.3
70	7.98	0.05	20.35	16.2	211.02	17.5
80	8.37	0.03	14.8	16	221.53	16.69
90	8.5	0.001	20.9	5.89	237.55	5.92

Table 2: Table for maximal velocity and rotational velocity with errors dependent of the galactic longitude l (Radius R).

As explained in the theoretical background, the rotational velocity can now be deduced using the formula 31. Figure 8 shows the measured values with the theoretically predicted values for the rotational velocity of baryonic mass following from Binney and Tremaine's formula for the surface mass density of a stellar, gaseous disk [4]:

$$v_d = 452 \sqrt{\frac{\text{kpc}}{R} \left(1 - \left(\frac{R}{2 \text{kpc}} + 1 \right) \exp \left(\frac{R}{2 \text{kpc}} \right) \right)} \text{ kms}^{-1} \quad (31)$$

The measured data first rise to a maximum at around 6 kpc and then fall. After 7 kpc there is a slight upward trend, but with a relatively high error. Figure 9 shows a more precise measurement of the rotation curve of the Milky Way by Clemens (1985). There can not a rising trend with $R > 7$ kpc be recognised. The measurements above 7 kpc are therefore less accurate as seen in the larger errorbars. This can be explained by interference. You can recognise in general a similar structure. Our maximum rotational velocity is 240 kms^{-1} , whereas Clemens' is around 230 kms^{-1} , with many measuring points above this value. The remaining values are also relatively close to those of figure 9, so it can be said that our measurements are in agreement with those of Clemens.

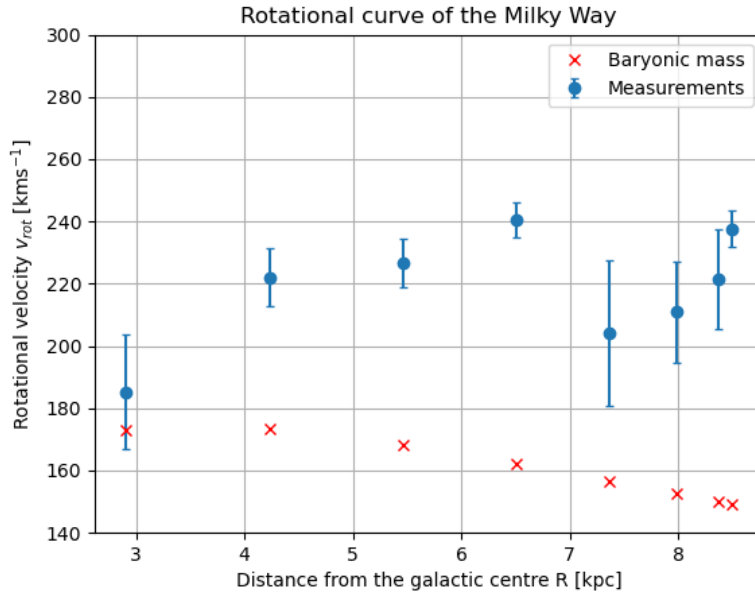


Figure 8: The plotted measurements illustrate the total and calculated baryonic velocities as a function of distance from the Galactic center. Detailed values and their corresponding errors are provided in Table 2.

2.4.3 Distribution of mass and dark matter in the Galaxy

The mass enclosed in the milky way from the centre of the galaxy to the distance R can be calculated as follows from [4]:

$$M_{<R} = 0.234 \cdot 10^{10} \left(\frac{v(R)}{100 \text{ kms}^{-1}} \right)^2 \left(\frac{R}{\text{kpc}} \right) M_{\odot}. \quad (32)$$

The baryonic matter distribution can therefore be derived from the velocity distribution. Thus, the dark matter distribution can be derived from:

$$M_{DM} = M - M_{\text{baryonic}}, \quad (33)$$

where M is the measured mass. The data is plotted in figure 10 and shown as a table in table 3. This difference between the theoretical assumption of the baryonic mass distribution and our

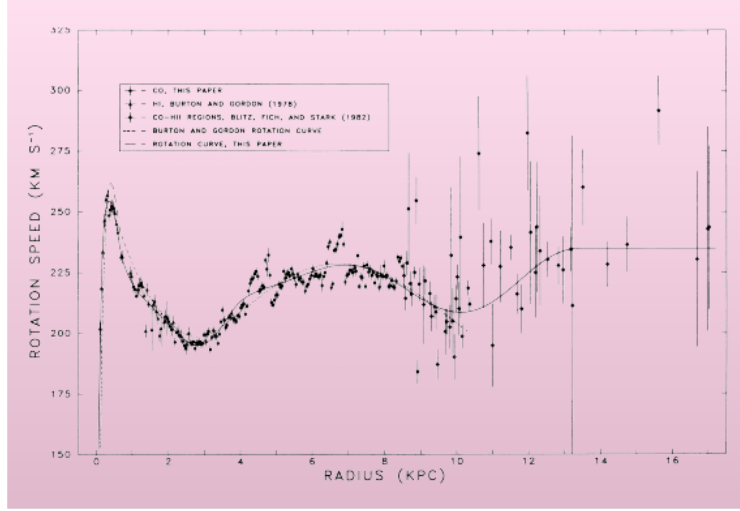


Figure 9: Rotation curve of the Milky Way. Figure taken from [4].

measurement points to two theories. Either mass does not behave as assumed in the theory, or there is more mass that cannot interact with matter, called dark matter.

Assuming that the rotational speed of the Milky Way remains constant after 8 kpc, the mass of the Milky Way can be calculated at a radius of 100 kpc. To do this, take the average of the last three rotation velocity values. With $v = 223.37 \pm 12.7 \text{ km s}^{-1}$ and equation 32, a value of $M(100 \text{ kpc}) = 1.16 \pm 0.14 \cdot 10^{12} M_{\odot}$ is obtained. This value is about a factor of 11 above the expected 100 billion solar masses for the stellar mass of the Milky Way. This suggests that the Milky Way consists in total of $8.6 \pm 0.9 \%$ baryonic matter and about $91 \pm 0.9 \%$ dark matter. For comparison, the total mass enclosed in the solar orbit is about 112 billion solar masses 3, which is about as much as the total stellar mass in the Milky Way. This supports the approximate ratio of 11 : 1 of dark matter mass and baryonic mass.

As can be seen in figure 10, dark matter is not as dense in the centre of the Milky Way as baryonic matter. Therefore, more dark matter must occur outside the centre to support the previous considerations. It is therefore possible that a dark matter halo is formed around and in the Milky Way.

The mean density of dark matter can now be calculated. A spherical distribution is assumed for this purpose, therefore the following applies:

$$M_{\text{DM}}(R) = \frac{4}{3}\pi\langle\rho_{\text{DM}}\rangle R^3 \quad \text{or} \quad \langle\rho_{\text{DM}}\rangle = \frac{3}{4} \frac{M_{\text{DM}}}{R^3}. \quad (34)$$

For $R = R_0 = 8.5 \text{ kpc}$, the solar orbit, you get $\langle\rho_{\text{DM}}\rangle = 1.78 \pm 0.17 \cdot 10^{-21} \text{ kg m}^{-3}$ as an approximate result. Furthermore, the dark matter energy density can be derived as $\langle\rho_{\text{DE}}\rangle = \langle\rho_{\text{DM}}\rangle \cdot c^2 = 1.00 \pm 0.09 \text{ GeV cm}^{-3}$, whereas $1 \text{ GeV cm}^{-3} = 1.6 \cdot 10^{-4} \text{ J m}^{-3}$.

Radius [kpc]	v_{rot} [kms $^{-1}$]	M_{total} [$10^{10}M_{\odot}$]	Error M_{total} [$10^{10}M_{\odot}$]	M_{baryonic} [$10^{10}M_{\odot}$]	M_{DM} [$10^{10}M_{\odot}$]
2.9	118.24	2.32	1.59	2.03	0.29
4.24	222.09	4.89	0.95	2.99	1.90
5.46	226.71	6.56	0.83	3.61	2.94
6.51	240.5	8.81	0.63	3.99	4.81
7.36	204.27	7.18	2.23	4.21	2.97
7.98	211.02	8.31	1.59	4.33	3.97
8.37	221.53	9.61	1.65	4.40	5.20
8.5	237.55	11.22	0.65	4.42	6.80

Table 3: Table of baryonic mass, dark matter and total mass distributed along the Milky Way for different radii.

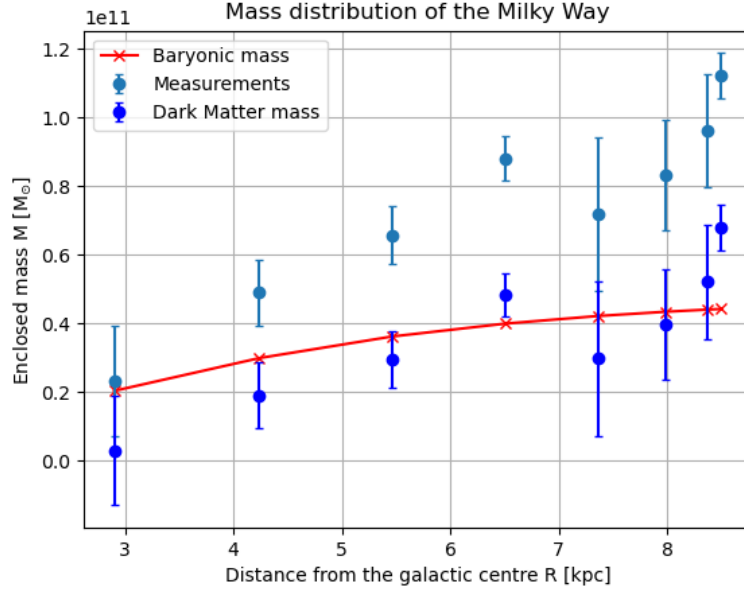


Figure 10: The distribution of mass as a function of radius is depicted, showcasing the distributions of total mass, baryonic mass, and dark matter of the Milky Way from 2.9 to 8.5 kpc.

3 Summary and discussion

3.1 Summary

In this chapter, the results of the observations and evaluation are summarized and put into context. The chronological structure of the previous section is used here.

Section 2.1 dealt with the calibration and calculated the system temperature of the left and right circular polarization as well as the measured and theoretical antenna temperature:

$$T'_{sys,left} = -1160.45 \text{ K} \pm 378.24 \text{ K}$$

$$T'_{sys,right} = 296.63 \text{ K} \pm 226.09 \text{ K}$$

$$T_{sun,left} = -2011.61 \text{ K} \pm 621.57 \text{ K}$$

$$T_{sun,right} = 2511.95 \text{ K} \pm 1141.88 \text{ K}$$

$$T_A = \frac{S_\nu A_e}{2k_B} = \frac{0.5\pi S_\nu (1.5)^2}{2k_B} = 896.38 \text{ K}$$

For the background radiation, the TPI and later the temperature was plotted as a function of azimuth and elevation. With an extrapolation the Temperature at 0° was determined:

$$T'_{sys}(0^\circ) = 257.13 \text{ K} \pm 18.14 \text{ K}$$

The angular resolution was determined in section 2.2. An average FWHM of $4.13^\circ \pm 0.43^\circ$ was determined. This value has an average deviation of 5.7 % from the literature value of 4.09° , which is accurate.

Section 2.3 examines the sensitivity of the instrument. First, the theoretical sensitivity in a single channel, the theoretical integration time for Taurus A and the signal-to-noise ratio were calculated:

$$\Delta S_\nu = 0.60 \cdot 10^{-23} \text{ W m}^{-2} \text{ Hz}^{-1} \pm 0.14 \cdot 10^{-23} \text{ W m}^{-2} \text{ Hz}^{-1}$$

$$\tau = 2.46 \text{ s} \pm 0.23 \text{ s}$$

$$\frac{S_\nu}{\Delta S_\nu} = 1.57 \pm 0.48$$

This was followed by the determination of the uncertainty of radiant flux measurements. First, the flux density of a cell phone for 900 MHz, the flux density of Cas A for 900 MHz and the antenna temperature with Cas A as the source were calculated:

$$S_{\nu,phone} = 1263.79 \text{ Jy}$$

$$S_{\nu,CasA} = 1954.37 \text{ Jy} \pm 98.31 \text{ Jy} = 1.55 \cdot S_{\nu,phone}$$

$$T_A = 1.82 \text{ K} \pm 0.09 \text{ K}$$

Finally, the antenna temperature of the source Tau A T_A , the signal to noise ratio S/N and the actual measurement time were calculated:

$$T_s = (5.66 \pm 465.27) \text{ K}$$

$$\frac{S}{N} \approx 1.02 \pm 3.9$$

$$\frac{(t_{tot} - t_{meas})}{t_{tot}} = 74.47 \%$$

In section 2.4, the 21 cm line of neutral hydrogen was measured and used to approximate the mass distribution in the Milky Way. To do this, the velocity resolution of the measuring instrument was first calculated, which is 12.9 km s^{-1} . The measurement data of the temperature of the antenna relative to v_{LSR} were then obtained, see figure 7. From this, the rotational velocity was calculated as a function of the distance to the galactic center as shown in figure 8. This was then finally used to calculate the mass distribution of hydrogen in the Milky Way in Figure 10. A discrepancy was found between the observed behavior of the masses and the expected behavior of baryonic mass. This was explained by dark matter. Some properties can be derived from this, including no interaction with electromagnetic radiation, but gravitational interaction. According to current knowledge, the Milky Way is surrounded by a dark matter halo.

3.2 Discussion

This chapter summarizes and discusses the corresponding sources of error. Once again, it is proceeded chronologically and a connection between the results is established.

The first and most important source of error in this experiment is the incorrect processing of the signal at the receiver on the left polarization side. Recognizable by the inappropriate temperature data, in particular $T'_{\text{sys},\text{left}}$, $T_{\text{sun},\text{left}}$ and all subsequent results, it must be a malfunction at the receiver. For this reason, the protocol only continues to work with the data from the right polarization side, as even a standard deviation can no longer explain the large errors. In the further course, the theoretically calculated antenna temperature of the sun T_A also shows a larger deviation and is not in the range of the measured values with the addition of their standard deviation, which could indicate that a larger background noise could have been permanently present during the measurement. A supervisor of the experiment considers that a loose plug or an insufficient connection of the telescope to the receiver could also be the problem. Nevertheless, part 2.1.2 shows a background temperature T_h by extrapolation for the right polarization side that coincides exactly with the earth radiation temperature T_{Earth} , which in turn suggests meaningful data for the right polarization side.

Sources of error in section (2.2) are, on the one hand, subsequent errors from the previous measurement that were used for normalization. On the other hand, there were difficulties with the right polarization of the latitude measurement in this measurement due to probable interference or technical problems. Another error is caused by weather events such as wind.

For the sensitivity determination of the instrument in section 2.3, further larger error propagation values can be seen in the theoretical calculation of the sensitivity, which in turn are the result of the deviations in the values for T'_{sys} from the previous calculations. The resulting error then continues in the further calculations, so that a high deviation in the calculated signal-to-noise ratio can also be seen. All these errors can also be attributed to the interference of the telescope and its receiver as well as noise. Later, when the maps are created, the sources of error are different. They are mainly characterized by unwanted radiation and interference, for example with the Milky Way. Radiation from the Earth also plays a role, as do other misprocessing by the telescope's receiver. Overall, however, the errors here are largely limited and the measurement data shown can be regarded as representative.

When evaluating the 21 cm line, the plots of T_A as a function of v_{LSR} were not all successful. The errors there extended over the entire evaluation of the HI lines. However, an indication of dark matter could still be given. Here too, the errors already mentioned, such as the excessive background noise and the technical interference at the telescope's receiver, play a role in explaining the difference between our measurements and the theoretical expectation. Furthermore, the rotation curve from the measured data can be compared with the literature values. It is noticeable that

the measured curve and the literature curve largely agree from 3 kpc to 7 kpc, but there are larger deviations from 7 kpc. This could be explained by possible interference that may have occurred during the measurement. In addition, the HI spectra can be compared with the structure of the Milky Way. It must be mentioned that the central region of the Milky Way, the "bulge region", which lies at $R < 3$ kpc, was not part of the measurements. This region is bar-shaped and hydrogen atoms do not move there on circular orbits as expected. Therefore, the calculations can only be an approximation of the real movements of the hydrogen atoms. Furthermore, it is noticeable that the total mass of the Milky Way increases linearly, although the rotation speed is approximately constant. This is an indication that there must still be "dark matter" in the Milky Way that is not in the visible range and does not interact with electromagnetic radiation. It is assumed that this dark matter is localized in a spherical halo around the Milky Way. The calculated curve of the mass of the dark matter is almost parallel to the curve of the calculated total mass and the mass of baryonic matter decreases with distance. From this it can be concluded that the density distribution of dark matter increases with increasing distance and that of baryonic matter decreases with increasing distance. This conclusion is consistent with the theoretical consideration that dark matter is distributed in a halo around the Milky Way.

In summary, it can therefore be said that the sources of error in the experiment were generally very high and initially poor values continued through further calculations in the course of the protocol. Nevertheless, by using two polarization sides, it was usually always possible to find a good side which provided good measurement data and which could be used for the calculations and evaluation. Thus, despite unfortunate technical disturbances, meaningful statements could be made and the main task of the experiment, the determination of the rotational velocity and the mass distribution of the Milky Way, could be fulfilled.

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Appendix

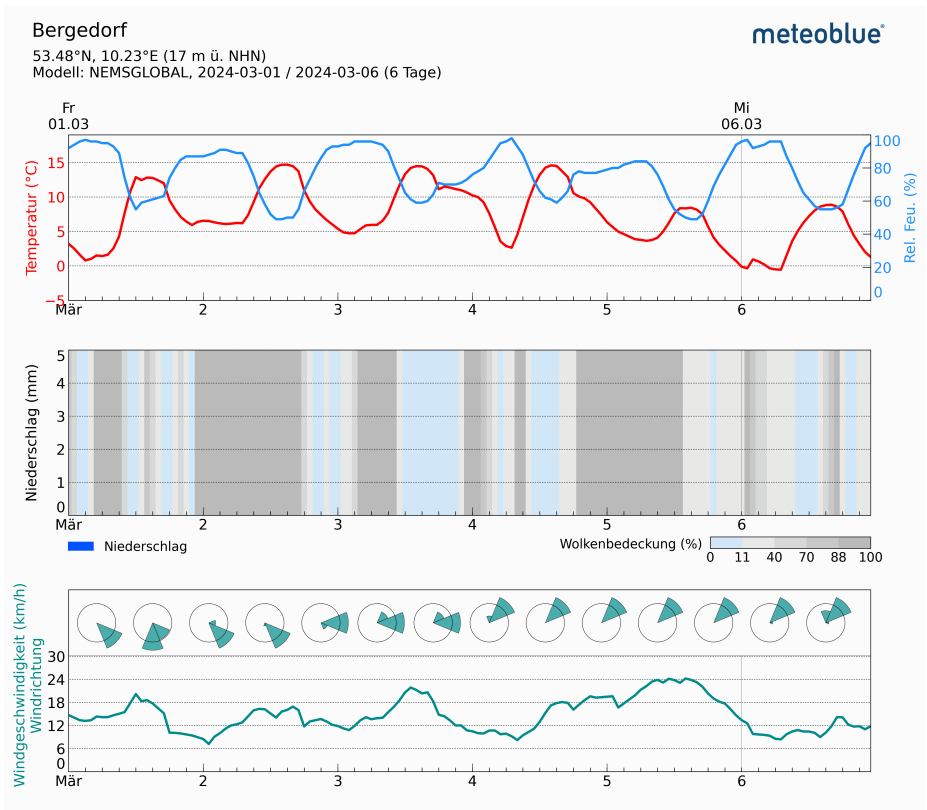


Figure 11: Weather data from March 1 to 6, 2024 [3].



Figure 12: Screenshot of the camera recording of the telescope in the program RadioUniversePRO on March 4, 2024 at 05:28 p.m..

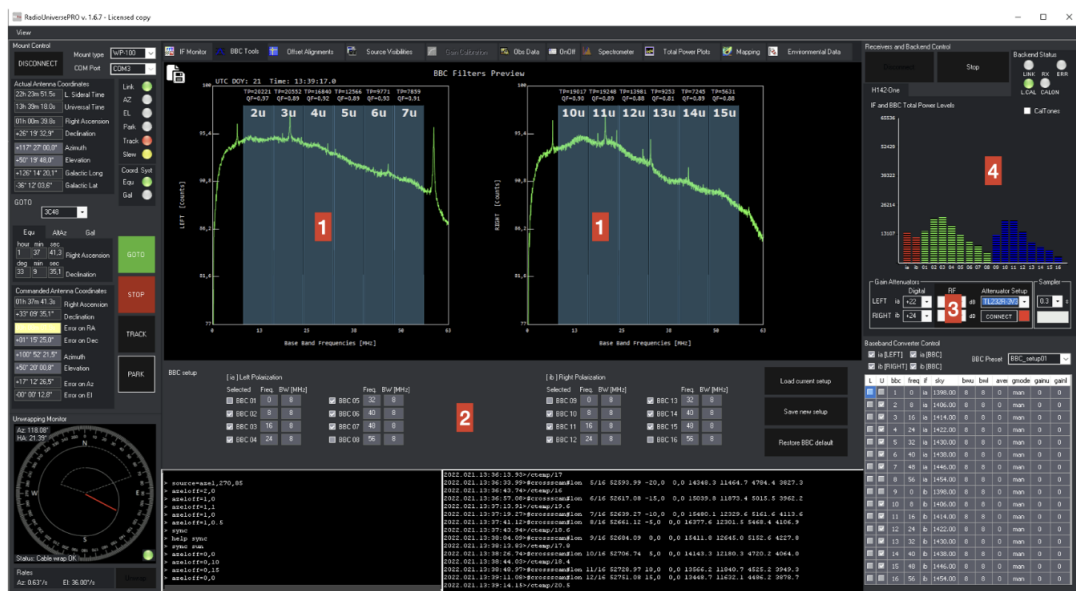


Figure 13: RadioUniversePRO user interface with the "BBC Tools" tab open. The following information is shown: uncalibrated spectra of left and right circular polarization (1), selection of frequency bands (2), receiver gain settings (3), bar graphs of the measured total power intensity in the frequency bands (4) [1].

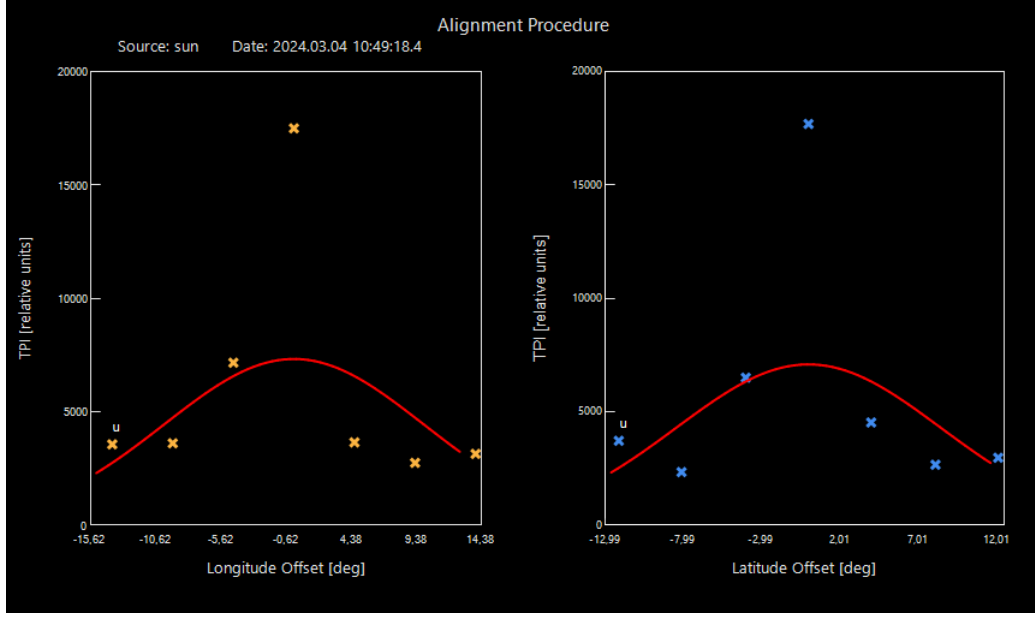


Figure 14: As an example, here is the first alignment procedure on March 4, 2024 at 10:49 a.m. The total power in counts as a function of longitude and latitude in degrees can be seen. A Gauss fit was automatically placed over the measurement points by the program.

BBC	P on	P off	P cal	T _{sys} ' [K]	T _{sun} [K]
1.0	12393.2	5014.6	4515.2	-1707.01	-1707.52
2.0	16291.4	5202.4	4442.4	-1163.69	-2473.52
3.0	15703.0	4669.8	4116.6	-1435.04	-2741.77
4.0	13618.8	5060.2	3949.6	-774.57	-1962.74
5.0	10631.0	5288.0	4042.8	-721.94	-1172.52
9.0	20789.2	4554.4	5615.0	730.01	1057.37
10.0	25033.2	3575.6	5852.4	266.98	1780.1
11.0	21064.2	1596.2	4040.2	111.03	3617.82
12.0	15887.6	1188.8	2848.8	121.74	3667.64
13.0	11850.6	1286.0	2148.8	253.38	2436.83

Figure 15: Measured and calculated data of P_{on} , P_{off} , P_{cal} , T'_{sys} , T_{sun} . The temperatures are in units of Kelvin the units of the measured powers are arbitrary.

Mean T_sys'_left [K]	Std T_sys'_left [K]	Mean T_sys'_right [K]	Std T_sys'_right [K]
-1160.45	378.24	296.63	226.097

Figure 16: Calculated means and standard deviations of $T'_{sys,left}$ and $T'_{sys,right}$.

Mean T_sun_left [K]	Std T_sun_left [K]	Mean T_sun_right [K]	Std T_sun_right [K]
-2011.61	621.57	2511.95	1141.88

Figure 17: Calculated means and standard deviations of $T_{sun,left}$ and $T_{sun,right}$.

Pointing settings		
Repetitions	:	1
Integration period [s]	:	2
Approx. duration [s]	:	28
Points per axis	:	7
Step size (HPBW frac)	:	1
Field of pointing [deg]	:	28,24

(a) Pointing settings for checking the pointing accuracy at the beginning of each lab day.

On-off- settings		
Repetitions	:	5
Average [s]	:	3
Elev cutoff [°]	:	60
Distance [beam]	:	4
Checkbox galoff	:	off
Checkbox Acquire OnOff Spectra	:	off
Checkbox Save as FITS	:	off

(b) On-Off settings for the measurement of the system temperature.

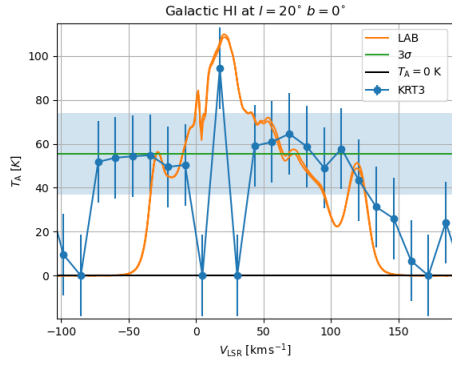
Total Power Plots settings		
Reference frame	=	Horizon
Span [deg]	=	68.48
Step [deg]	=	5
Average [s]	=	3
Devices	:	BBCs
Checkbox Save as FITS	:	OK

(c) Total Power Plots settings for the cross-scan for the measurement of the background radiation.

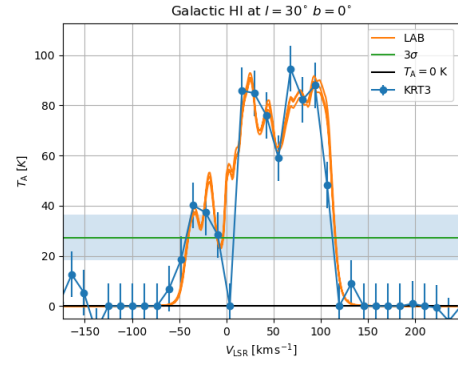
Mapping settings		
Width	:	10°
Height	:	10°
Delta	:	0.5
Average	:	10s

(d) Mapping settings for the determination of the uncertainty of radiant flux measurements.

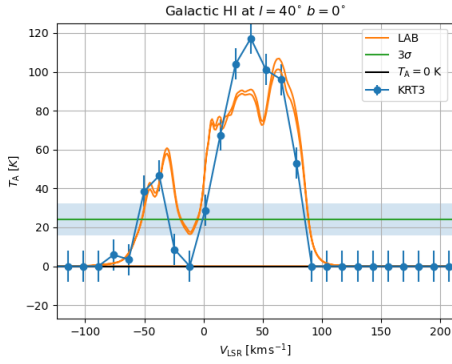
Figure 18: Handwritten log book entries.



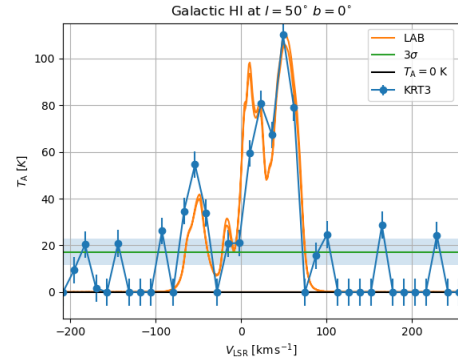
(a)



(b)

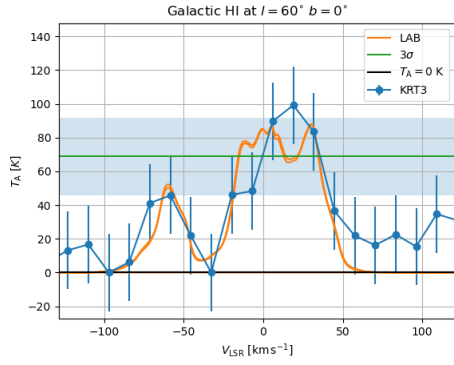


(c)

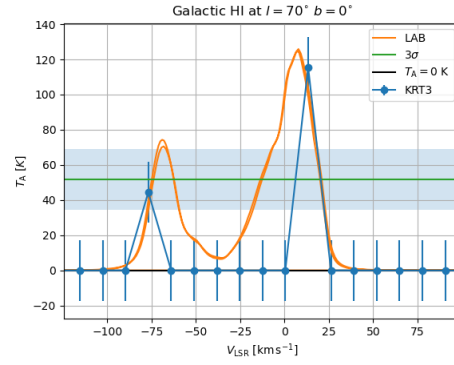


(d)

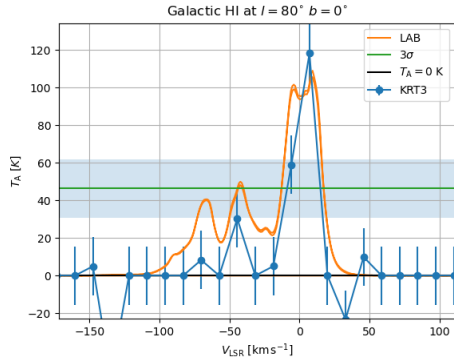
Figure 19: Plots of the spectrum of the HI 21 cm line at galactic longitude from 20° to 50° .



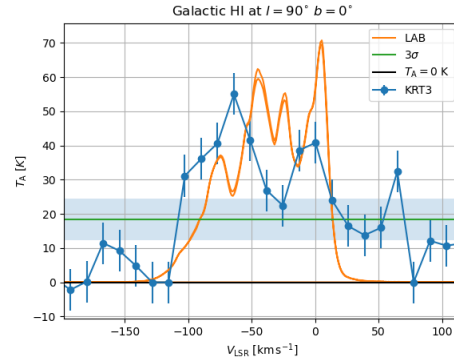
(a)



(b)



(c)



(d)

Figure 20: Plots of the spectrum of the HI 21 cm line at galactic longitude from 60° to 90° .

4.3.24: 9,3°C, bewölkt 6/8

- Die Bänder 6,7,8, 14,15,16 sind rausgenommen worden wegen techn. Schwierigkeiten (unter Perena)
- Punktung-korrektur war nicht nötig

SWC → $S_{V=14,20,44,112}$

$$S_v = \frac{T_A}{A_e} \cdot \epsilon_{ho} \quad \leftarrow \text{Antennentemp.}$$

↖ rechnerisch

$$\left[\begin{array}{l} T'_{sys} = T_{cal} \frac{P_{off}}{P_{cal} - P_{off}} \\ T'_{sys} = T_{sun} \frac{P_{off}}{P_{on} - P_{off}} \end{array} \right], T_{cal} = 170 \text{ K}$$

$$\frac{\omega}{m^2 H_z}$$

$$T'_{sys, cal} = \frac{\omega}{m^2 H_z}$$

~~Cyg A wurde ausgewählt (EC = 42,5°)~~

• Tau A (37,5°) (elevation 3,8°)

(T'_{sys} aus 5.1.1), $\tau = 31.02$ (integration)

10

5.4 (point correction error at 5° offset long.)

$$\frac{\Delta v}{v_b} = \frac{\Delta v}{c}$$

Figure 21: The notes taken while the experiment.